

Outline_COT / MEOW metrics 的 N 與 detectable difference 曲線報告

Codex + OMX xhigh assisted research

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Executive summary

本報告回答：以 survey/review paper 為樣本單位，在現行 Outline_COT / MEOW-style evaluation metrics 下，比較兩個 model、prompt 或 run 時，多少 paper 數量 N 才可能檢出統計上顯著的差距，以及給定 N 時至少要多大的分數差距。主設定是 paired comparison、two-sided $\alpha = 0.05$ 、power = 0.8，並附 power = 0.9 與 multiple-comparison sensitivity。

結論很直接：N=4 只能作 smoke / exploratory signal。對 paired continuous 近似，N=4 在 80% power 下需要約 $d_z = 1.40$ 的標準化效果；N=100 需要約 $d_z = 0.28$ 。如果 metric 的 paired 差值標準差 σ_d 很大，raw score 的門檻會按比例變大。

Metric inventory

Metric	Source / output	Range / direction	Statistical treatment
Structural Distance	scripts/combine_scores.py; normalized scripts/evaluate_chatgpt_metrics/5D-blind-watch-distance/result.structural_distance	0-10, lower better	Paired bounded continuous plus permutation/bootstrap.
6D judge dimensions	prompts/meow_llm_judge_6D-source-higher.txt; SCORE_KEYS; re- result.judge_scores.*	0-10, higher better	Ordinal bounded scores; continuous approximation for planning; ordinal/permutation robustness.
Judge average	Mean of six dimensions in result summaries	0-10, higher better	Composite bounded continuous; dimension correlation means it is not six independent tests.
Reference reward	scripts/ref_reward.py; scripts/evaluate_pair_rewards.py	auxiliary reward, higher better	Excluded from main curves per current scope; include only if promoted to active output.

Structural Distance 是 shape-only：本地樹建構只使用 section level，節點標籤固定，故 title wording、numbering、references 與語義內容不會影響此 metric。Local judge 是 upstream repo-defined 6D judge，六維為：结构 _ 信息快速定位, 结构 _ 详略得当, 内容 _ 章节互斥性, 内容 _ 逻辑深度, 内容 _ 学术价值, 语用 _ 描述性与简洁性。

Mathematical core

令同一篇 paper 在兩個條件下的差值為 $d_i = y_{B,i} - y_{A,i}$ ，paired 差值標準差為 σ_d 。常態近似下：

$$\Delta_{min}(N) = (z_{1-\alpha/2} + z_{power}) \frac{\sigma_d}{\sqrt{N}}$$
$$N_{min}(\Delta) = \left(\frac{(z_{1-\alpha/2} + z_{power}) \sigma_d}{\Delta} \right)^2$$

在 $\alpha = 0.05$ 、power=0.8 時，係數為 2.802；power=0.9 時為 3.242。小樣本下這是 optimistic planning approximation，正式結論應用 paired permutation / bootstrap confidence intervals 驗證。

Standardized detectable effect

N papers	d_z power 0.8	d_z power 0.9
4	1.401	1.621
10	0.886	1.025
25	0.560	0.648
50	0.396	0.458
100	0.280	0.324
200	0.198	0.229

Raw detectable deltas: 0–1 structural-like metrics

N	sigma_d=0.05	sigma_d=0.10	sigma_d=0.20
4	0.070	0.140	0.280
10	0.044	0.089	0.177
25	0.028	0.056	0.112
50	0.020	0.040	0.079
100	0.014	0.028	0.056
200	0.010	0.020	0.040

Raw detectable deltas: 0–10 judge-like metrics

N	sigma_d=0.5	sigma_d=1.0	sigma_d=1.5
4	0.700	1.401	2.101
10	0.443	0.886	1.329
25	0.280	0.560	0.840
50	0.198	0.396	0.594
100	0.140	0.280	0.420
200	0.099	0.198	0.297

Empirical paired calibration

現有 local artifacts 中，真正可用的 paired overlap 多為 N=3 或 N=4，因此以下只作曲線校準，不能作穩定母體估計。工作樹目前也存在大量既有 modified/untracked result files，故 artifact provenance 需要在正式引用前再次固定。另需注意：gemini-3.1-pro-preview-3papers 的 judge model 在現有 artifact 中不完全一致，本報告不把它用作乾淨 judge-score variance 校準。

Pair		Metric	N	Mean diff B-A	SD(diff)
ablation_with_meta_abstract vs ablation_no_meta_abstract		structural_distance	4	-0.082	0.048
ablation_with_meta_abstract vs ablation_no_meta_abstract		judge_average	4	0.167	1.835
gpt-5.4-xhigh-3papers vs gpt-5.4-mini-xhigh-4papers		structural_distance	3	-0.016	0.060
gpt-5.4-xhigh-3papers vs gpt-5.4-mini-xhigh-4papers		judge_average	3	-0.250	0.520
gpt-5.4-xhigh__judge-gemini-3.1-pro-preview vs gpt-5.4-xhigh-3papers		structural_distance	3	0.000	0.000
gpt-5.4-xhigh__judge-gemini-3.1-pro-preview vs gpt-5.4-xhigh-3papers		judge_average	3	0.389	1.088

Figures

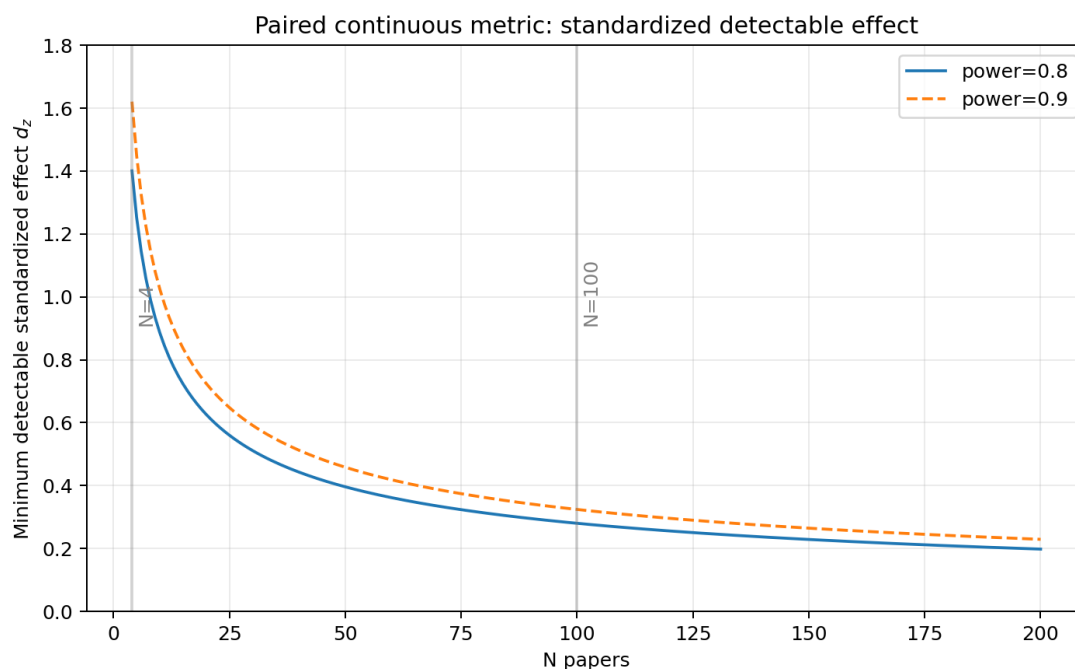


Figure 1: Paired continuous metric 的標準化 detectable effect 。

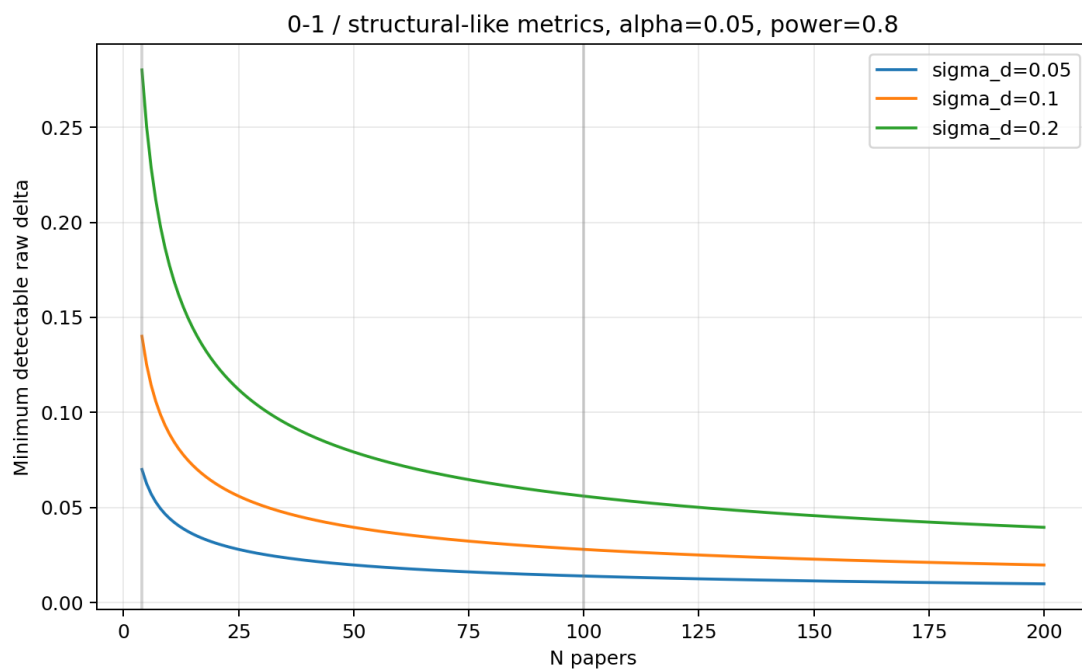


Figure 2: 0-1 / Structural-like metric raw delta curve °

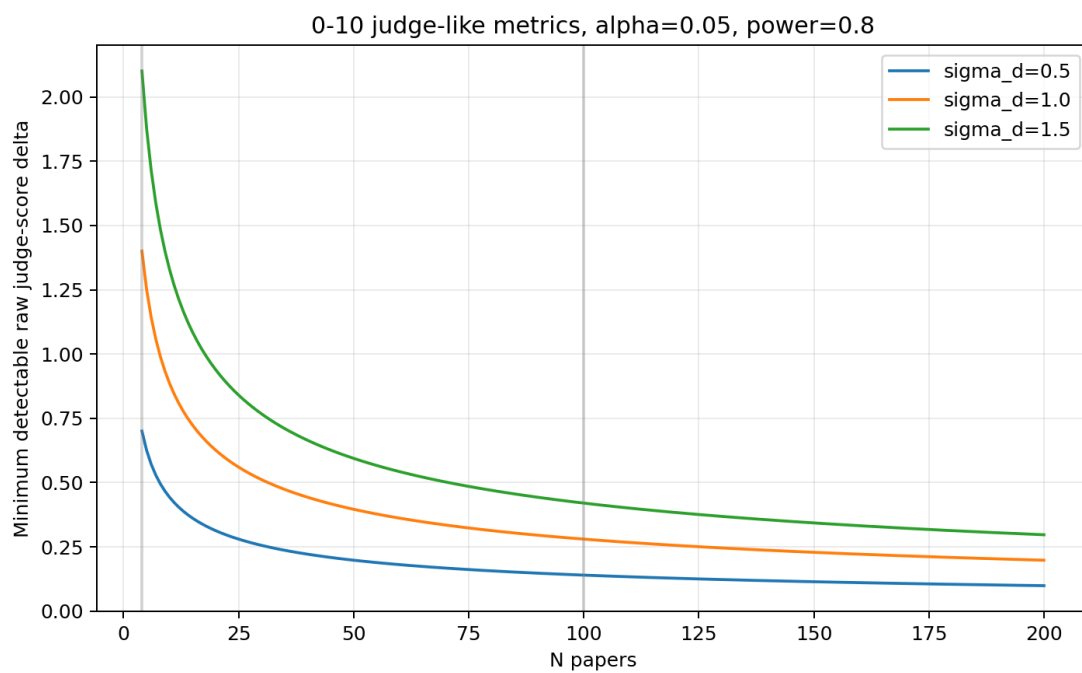


Figure 3: 0-10 judge-like metric raw delta curve °

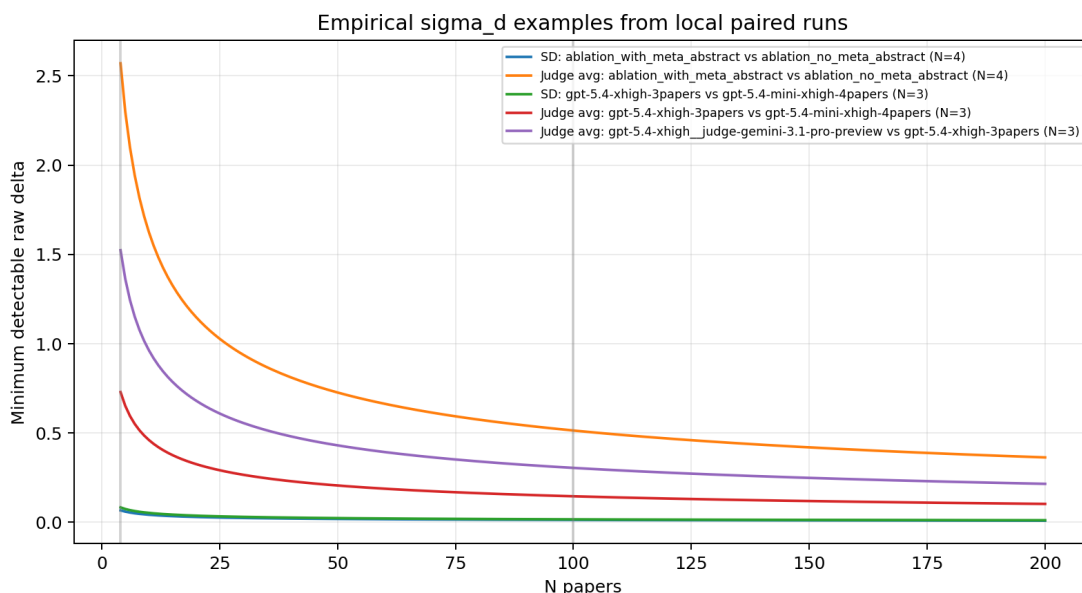


Figure 4: Local empirical paired sigma examples ◦

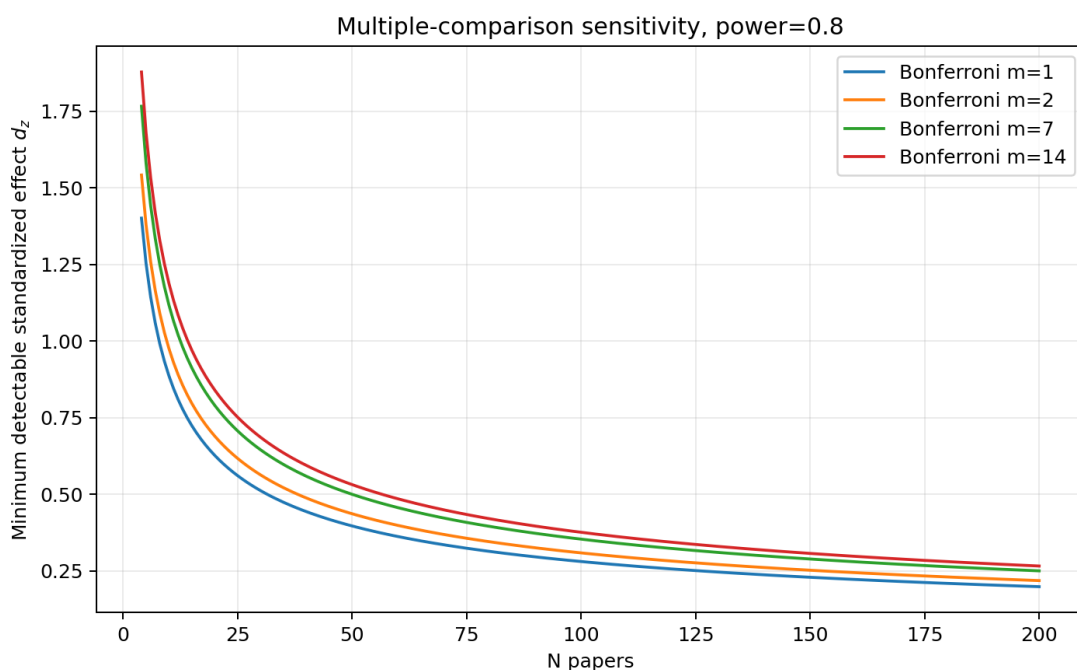


Figure 5: Bonferroni multiple-comparison sensitivity ◦

Recommendations

- $N=4$ ：只當 smoke。除非 raw gap 大到約 $1.4\sigma_d$ ，否則不應期待穩定顯著。
- $N=25$ ：可檢出約 $0.56\sigma_d$ ，適合 early benchmark / ablation triage。
- $N=50$ ：可檢出約 $0.40\sigma_d$ ，開始接近中等效果的正式比較。
- $N=100$ ：可檢出約 $0.28\sigma_d$ ，適合主張 moderate but meaningful improvement。
- 對 6D judge，不要把六維當成六個完全獨立樣本；若逐維報告，使用 Holm 或 Bonferroni，若作探索則用 BH/FDR。

- 對 Structural Distance，先確定研究問題真的是 shape alignment；它不評語義、引用、標題品質。

Limitations

本報告沒有重新跑 judge，也沒有把 auxiliary Reference reward 納入主曲線。Empirical sigma 只來自現有 local artifacts，且 N 很小，因此報告主體應視為 mathematical planning + source-confirmed metric inventory，而不是最終實驗結果。